

RESEARCH SKILLS

PNAP0164

WEEK 3 :
THE HEART OF RESEARCH PROCESS

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LEARNING OUTCOMES

By the end of this lesson, you will be able to:

1. **Explain** the central role of problem definition in research.
2. **Apply** techniques to define a clear, researchable problem.
3. **Formulate** research questions, objectives, and hypotheses.

INTRODUCTION

The heart of research

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Every research begins with identifying a meaningful problem. A well-defined research problem provides direction for the entire study, including the research design, methodology, data collection, and analysis. Without a clear problem, research may become unfocused, inefficient, and unable to produce meaningful findings.

Key Points

- The research problem is the foundation of every study.
- It determines the direction of the research.
- It guides the selection of appropriate methods.
- It ensures the study addresses a real need.

This stage **sets the direction** for design, methodology, and analysis

DEFINING THE RESEARCH PROBLEM

Definition

“A research problem is an issue, concern, or gap in knowledge that requires systematic investigation to produce new understanding or solutions.”
(Kumar, 2023)

Key Features

- Clear and specific
- Researchable using data
- Relevant to the field
- Significant to society or academia
- Feasible within available time and resources

NECESSITY OF DEFINING THE PROBLEM

Focus :

Keeps the research on a specific issue and prevents unnecessary expansion.

Direction :

Guides the selection of research design, methodology, and data collection.

Contribution :

Increases the potential of the study to generate new knowledge or practical solutions.

Relevance :

Ensures the study addresses a meaningful academic or real-world issue.

Feasibility :

Helps researchers complete the study within available time, budget, and resources.

1

Literature Review :

Identify unanswered questions, inconsistencies, or research gaps.

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Current Issues :

Examine emerging trends, government policies, technological developments, or societal challenges.

TECHNIQUES FOR DEFINING A PROBLEM

2

Observation :

Observe issues, behaviours, or situations occurring in the real world.

3

Personal experience:

Reflect on problems encountered during study, work, or daily life.

4

Consult Experts :

Discuss ideas with supervisors, lecturers, researchers, or practitioners.

5

Gap Analysis :

Compare existing knowledge with what still needs to be explored.

FROM PROBLEM TO RESEARCH

A clear alignment of each component ensures your research is focused, logical, and meaningful.



RESEARCH GAP

**“A missing or underexplored area of knowledge that justifies the need for a new study.”
(Kumar, 2023; Creswell & Creswell, 2023)**



Abstract 1

"Many studies have explored the use of e-learning platforms among university students. However, limited research has examined how gifted secondary school students adapt to project-based learning through AR/VR technologies in hybrid classrooms."

Find :

1. What has been studied
2. What is missing?
3. What is the research gap?

Abstract 2

"Existing research shows that technopreneurship education improves university students' entrepreneurial intentions. However, little is known about how early exposure to digital entrepreneurship bootcamps influences innovative thinking among secondary school students."

Find :

1. What has been studied
2. What is missing?
3. What is the research gap?

Abstract 3

"Numerous studies have examined the impact of digital learning platforms on academic performance. However, less attention has been given to how extended screen time in digital learning affects the mental well-being and social skills of secondary school students."

Find :

1. What has been studied
2. What is missing?
3. What is the research gap?

Abstract 4

"Previous research highlights that peer mentoring programs improve collaboration and academic outcomes at the university level. Yet, there is little research exploring how peer mentoring can foster leadership and teamwork skills in gifted secondary school students."

Find :

1. What has been studied
2. What is missing?
3. What is the research gap?

Abstract 5

"Research on smart classroom technologies has primarily focused on their technical efficiency and adoption rates among teachers. However, fewer studies have explored how smart classroom technologies influence creativity and problem-solving skills among students in hybrid learning environments."

Find :

1. What has been studied
2. What is missing?
3. What is the research gap?

RESEARCH GAPS VS RESEARCH PROBLEMS

Aspect	Research Gap	Research Problem
Definition	A research gap is the missing piece of knowledge in existing studies. It is something that has not yet been studied, or not studied enough.	A research problem is the issue, difficulty, or challenge that needs to be addressed through research. It often comes from the gap, but expressed as a problem statement.
Key Difference	The missing knowledge in the literature	The practical or theoretical issue caused by that gap, which your research will solve.
Source	Found through literature review	Derived from the gap, linked to real-world issue
Purpose	Justifies why new research is needed	Defines what your research will actually solve
Example	“Few studies on AR/VR for gifted students.”	“Educators lack strategies to use AR/VR for gifted students.”



RESEARCH QUESTIONS

A good research question is clear, focused, and researchable (O'Leary, 2021).

Types:

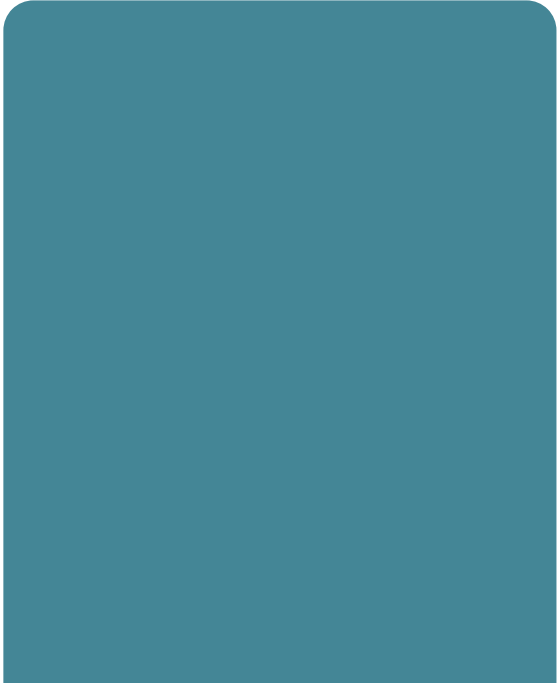
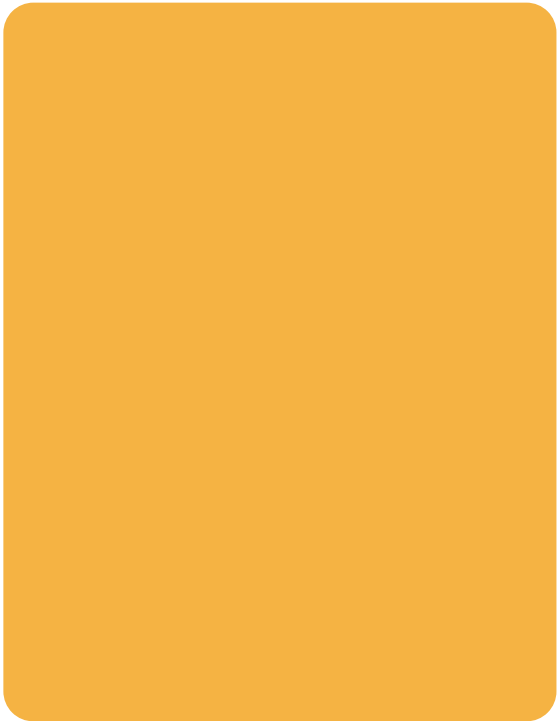
- Descriptive (“What is...?”)
- Comparative (“What is the difference between...?”)
- Relational (“Is there a relationship between...?”)

Example:

“How does AR-based PBL influence problem-solving skills in gifted students?”

RESEARCH PROBLEMS VS RESEARCH QUESTIONS

Aspect	Research Problem	Research Questions
Definition	The issue or challenge that needs to be addressed.	The specific questions your study will answer in order to address the research problem.
Key Difference	The “big issue” that needs solving.	The “small, answerable steps” that break the problem into pieces.
Nature	Broad, general statement of what is wrong, missing, or unclear.	Narrow, focused, and researchable.
Purpose	To justify why the study is necessary.	To guide the data collection, analysis, and findings.
Example	"Despite the rapid adoption of AR/VR in education, little is known about how gifted secondary school students adapt to project-based learning in hybrid classrooms."	• How do gifted secondary school students adapt to AR/VR-based project learning in hybrid classrooms? • What challenges do they face? • What strategies can enhance their learning experience?



RESEARCH OBJECTIVES

General Objective:

Overall purpose of the research

Specific Objectives:

Measurable steps to achieve the general objective

SMART Framework:

- Specific
- Measurable
- Achievable
- Relevant
- Time-bound

Every research objective should correspond to at least one research question.

HYPOTHESIS

Definition: A testable prediction about the relationship between two or more variables.

Types:

- Null Hypothesis (H_0) – no relationship or effect.
- Alternative Hypothesis (H_1) – predicts a relationship or effect.
- Directional vs Non-directional.

Example:

- H_0 : AR-based PBL has no effect on problem-solving skills.
- H_1 : AR-based PBL improves problem-solving skills.

Note: Qualitative studies generally do not formulate hypotheses.

RESEARCH SCOPE

DEFINITION

- The boundaries of the study — what the research will cover and what it will not.
- Ensures the study remains focused, feasible, and realistic (Leedy & Ormrod, 2021).

KEY ELEMENTS

- **Population/Sample** – Who will be studied.
- **Geographical Area** – Where the study will take place.
- **Timeframe** – Period of study.
- **Variables/Themes** – What concepts or relationships are investigated.

Tip: Clearly stating the scope avoids overgeneralisation and aligns expectations.

CASE STUDY

TOPIC

A Technology-based Technopreneurship Education
Framework: Project-Based Learning Approach for
Gifted Students in the Sustainable Digital Economy

Research Problem:

Gifted students lack targeted technopreneurship training aligned with sustainability goals.

Research Question:

How effective is a technology-based PBL framework in enhancing entrepreneurial competencies of gifted students?

Research Objective:

To design and evaluate a technology-based PBL curriculum integrating SDGs.

Hypothesis:

Students in the PBL group will show greater improvement in entrepreneurial competencies compared to those in traditional instruction.

Research Scope:

- Sample: 40 gifted students in a Digital Innovator Programme
- Location: Secondary school gifted education in Malaysia
- Timeframe: 12 weeks
- Variables: Entrepreneurial competencies, sustainability awareness, and student perceptions

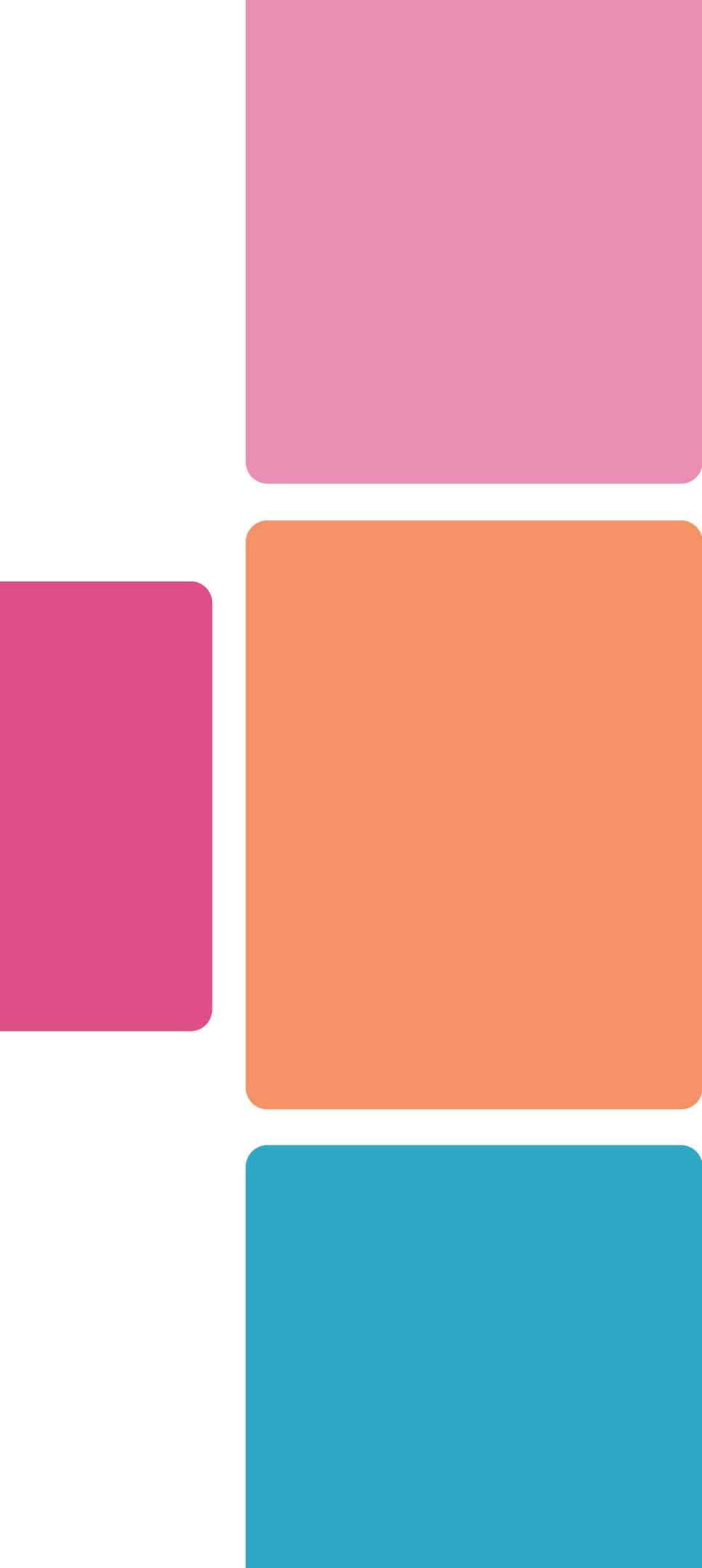
SUMMARY & DISCUSSION PROMPT

SUMMARY :

- Defining a problem is the heart of the research process.
- A strong research foundation includes:
 - Clear Problem Statement
 - Research Questions
 - Objectives
 - Hypotheses
 - Scope (boundaries)
- All must align logically to create a valid, reliable, and feasible study.

DISCUSSION PROMPT :

1. Why is it important to define the scope early in research?
2. Take a topic of interest → draft a short problem statement and identify its scope.
3. Which part is most challenging for you: Problem, RQ, RO, Hypothesis, or Scope?



THANK YOU